



US012410090B2

(12) **United States Patent**
Gross et al.

(10) **Patent No.:** **US 12,410,090 B2**

(45) **Date of Patent:** **Sep. 9, 2025**

(54) **NON-IOX GLASSES WITH HIGH
COEFFICIENT OF THERMAL EXPANSION
AND PREFERENTIAL FRACTURE
BEHAVIOR FOR THERMAL TEMPERING**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/203,890**

(22) Filed: **May 31, 2023**

(65) **Prior Publication Data**

US 2023/0312395 A1 Oct. 5, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/767,191, filed as
application No. PCT/US2018/063404 on Nov. 30,
2018, now Pat. No. 11,708,296.

(60) Provisional application No. 62/592,683, filed on Nov.
30, 2017.

(51) **Int. Cl.**
C03C 3/097 (2006.01)
C03B 27/012 (2006.01)
C03C 3/091 (2006.01)

(52) **U.S. Cl.**
CPC **C03C 3/097** (2013.01); **C03B 27/012**
(2013.01); **C03C 3/091** (2013.01)

(58) **Field of Classification Search**
CPC C03C 3/097; C03C 3/091
See application file for complete search history.

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(57) **ABSTRACT**

The disclosure relates to glass compositions with high
coefficients of thermal expansion and low fracture toughness
designed for thermal tempering. These glasses are ideally
suited to produce a "dicing" pattern when thermally tem-
pered, even when thin (<3 mm). Disclosed glasses have high
thermal expansions at low and high temperatures to produce
increased temper stresses once quenched, coupled with low
fracture toughness which promotes crack bifurcation and
enhanced fragility. Methods of making such glasses are
also provided.

20 Claims, No Drawings

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NON-IOX GLASSES WITH HIGH COEFFICIENT OF THERMAL EXPANSION AND PREFERENTIAL FRACTURE BEHAVIOR FOR THERMAL TEMPERING

This application is a continuation of U.S. application Ser. No. 16/767,191 filed May 27, 2020, which claims the benefit of priority under 35 U.S.C. § 371 of International Application No. PCT/US18/063404, filed on Nov. 30, 2018, which claims the benefit of priority under 35 U.S.C. § 120 of U.S. Application Ser. No. 62/592,683 filed on Nov. 30, 2017, the content of which is relied upon and incorporated herein by reference in its entirety.

FIELD OF DISCLOSURE

The disclosure relates to fusion formable glasses with high coefficients of thermal expansion and low fracture toughness designed for thermal tempering. The glasses are ideally suited to produce a “dicing” pattern when thermally tempered, even at very thin thicknesses. They have high thermal expansions at low and high temperatures to produce increased temper stresses once quenched, coupled with low fracture toughness which promotes crack bifurcation and enhanced frangibility.

BACKGROUND

Thermally-tempered glass, sometimes called safety glass, is often applied in situations where safe fracture behavior is required to prevent injury in the case of failure. For example, safety glass is used to strengthen automobile side and rear windows, as well as objects such as shower doors. The property of tempered glass that makes it so desirable in safety applications is that when it breaks, it shatters into rock salt-like, cubicle pieces which do not have sharp edges or needle-like points. This desired fracture behavior is called “dicing” and occurs when the glass has achieved full temper.

In addition to the safety aspect of thermally tempered glass, tempering strengthens the glass, making it more damage resistant and durable. Because of the increased durability, tempered glass can be used in applications where normal glass would quickly break—for example, automotive windows, where the glass may be impacted by rocks or other hard materials. Due to the increase in glass use in architectural, automotive, and electronic device applications, there is a continuing demand for thin, thermally strengthened glasses that are durable, but when broken, break in a ‘safe’ or dicing break pattern. However, as glass becomes thinner it becomes harder to produce any thermal tempering stresses at all, and the central tension required for a safe “dicing” fracture pattern increases—producing a compound challenge. The present disclosure solves this unmet need by disclosing glasses that produce enhanced temper stresses while needing to reach a lower stress level for dicing to occur.

SUMMARY

In an aspect (1), the disclosure provides a glass composition comprising greater than 70 mol % SiO_2 , greater than 0 to 20 mol % Al_2O_3 , greater than 0 to 20 mol % of at least one alkaline earth, oxide from the group consisting of MgO , CaO , BaO , or SrO , 3-16 mol % K_2O , >0-10 mol % B_2O_3 . In an aspect (2), the disclosure provides the glass composition of aspect (1), comprising 72-90 mol % SiO_2 . In an aspect (3), the disclosure provides the glass composition of

aspect (1) or aspect (2), further comprising 0-16 mol % Na_2O and 3-22 mol % $\text{Na}_2\text{O}+\text{K}_2\text{O}$. In an aspect (4), the disclosure provides the glass composition of any of aspect (1)-(3), wherein the low temperature coefficient of thermal expansion from 25-300° C. is greater than 75×10^{-7} ppm/° C. In an aspect (5), the disclosure provides the glass composition of any of aspect (1)-(4), wherein the high temperature coefficient of thermal expansion is greater than 250×10^{-7} ppm/° C. In an aspect (6), the disclosure provides the glass composition of any of aspect (1)-(5), wherein the fracture toughness, KIC is less than 0.65 MPa-m^{1/2}. In an aspect (7), the disclosure provides the glass composition of any of aspect (1)-(6), wherein the glass has a viscosity of 200 kP at a temperature of from 1000-1200° C. In an aspect (8), the disclosure provides the glass composition of any of aspect (1)-(7), wherein the glass has a viscosity of 35 kP at a temperature of from 1100-1300° C. In an aspect (9), the disclosure provides the glass composition of any of aspect (1)-(8), comprising 0.5 to 8 mol % or 1 to 5 mol % B_2O_3 . In an aspect (10), the disclosure provides the glass composition of any of aspect (1)-(9), comprising >0-10 mol % or 3 to 10 mol % Al_2O_3 . In an aspect (11), the disclosure provides the glass composition of any of aspect (1)-(10), comprising 2 to 20 mol % or 2 to 15 mol % MO . In an aspect (12), the disclosure provides the glass composition of any of aspect (1)-(11), comprising 8-16 mol % $\text{Na}_2\text{O}+\text{K}_2\text{O}$. In an aspect (13), the disclosure provides the glass composition of any of aspect (1)-(12), comprising 5-15 mol % K_2O . In an aspect (14), the disclosure provides the glass composition of any of aspects (1)-(13), wherein the glass is thermally tempered and has a depth of compression of at least about 10 microns. In an aspect (15), the disclosure provides the glass composition of any of aspects (1)-(14), wherein the glass is thermally tempered and has a surface compressive stress of at least 250 MPa. In an aspect (16), the disclosure provides the glass composition of any of aspects (1)-(15), wherein the glass meets the dicing standard ASTM C1048.

In an aspect (17), the disclosure provides a method of making the glass composition of any of aspects (1)-(16), the method comprising mixing SiO_2 , Al_2O_3 , at least one alkaline earth oxide from the group consisting of MgO , CaO , BaO , or SrO , K_2O ; and B_2O_3 to create a homogenous melt. In an aspect (18), the disclosure provides the method of aspect (17), wherein the melt is formed into glass sheets and the glass sheets are subsequently thermally tempered. In an aspect (19), the disclosure provides the method of aspect (17) or aspect (18), wherein the glass composition is meets the dicing standard ASTM C1048. In an aspect (20), the disclosure provides the method of any of aspects (17)-(19), wherein the glass is melted and maintained at temperatures ranging from 1100-1650° C. for times ranging from about 6-16 hours, and annealed at about 500-650° C., where the glass is held at temperature for about 1 hour and subsequently cooled for at least 6 hours.

DETAILED DESCRIPTION

In the following description, whenever a group is described as comprising at least one of a group of elements and combinations thereof, it is understood that the group may comprise, consist essentially of, or consist of any number of those elements recited, either individually or in combination with each other. Similarly, whenever a group is described as consisting of at least one of a group of elements or combinations thereof, it is understood that the group may consist of any number of those elements recited, either individually or in combination with each other. Unless otherwise speci-

fied, a range of values, when recited, includes both the upper and lower limits of the range as well as any ranges therebetween. As used herein, the indefinite articles “a,” “an,” and the corresponding definite article “the” mean “at least one” or “one or more,” unless otherwise specified. It also is understood that the various features disclosed in the specification and the drawings can be used in any and all combinations.

Where a range of numerical values is recited herein, comprising upper and lower values, unless otherwise stated in specific circumstances, the range is intended to include the endpoints thereof, and all integers and fractions within the range. It is not intended that the scope of the claims be limited to the specific values recited when defining a range. Further, when an amount, concentration, or other value or parameter is given as a range, one or more preferred ranges or a list of upper preferable values and lower preferable values, this is to be understood as specifically disclosing all ranges formed from any pair of any upper range limit or preferred value and any lower range limit or preferred value, regardless of whether such pairs are separately disclosed. Finally, when the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. When a numerical value or end-point of a range does not recite “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.”

As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. It is noted that the terms “substantially” may be utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. These terms are also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue. Thus, a glass that is “free of Al_2O_3 ” is one in which Al_2O_3 is not actively added or batched into the glass, but may be present in very small amounts as a contaminant (e.g., 500, 400, 300, 200, or 100 parts per million (ppm) or less or).

Unless otherwise specified, all compositions are expressed in terms of mole percent (mol %). Coefficients of thermal expansion (CTE) are expressed in terms of $10^{-7}/^\circ\text{C}$., unless otherwise specified. The low temperature CTE (LTCTE) is measured over a temperature range from 25°C . to 300°C . and expressed in terms of $10^{-7}/^\circ\text{C}$. The high temperature CTE (HTCTE) is measured at the temperature above glass transition region and expressed in terms of $10^{-7}/^\circ\text{C}$. The sum of the LTCTE and the HTCTE is expressed in terms of $10^{-7}/^\circ\text{C}$. The density in terms of grams/cm^3 was measured via the Archimedes method (ASTM C693). Young’s modulus, shear modulus, and Poisson’s Ratio were measured via the ASTM C623 standard.

Glass Compositions

In thermal tempering, a glass product is heated to near the softening temperature and then rapidly quenched, for example, by blowing cold air on the surfaces. As a result, the glass will possess a lower surface temperature than the interior during cooling. As the center of the glass cools more

slowly to room temperature it contracts to a smaller specific volume while the high specific volume of the surface layer remains unchanged. This leads to a surface compressive layer that gives tempered glass its strength. The difference in specific volume is due to a combination of differences in the thermal expansion of the glass upon cooling and from a fictive temperature difference between the surface and the bulk. To a first approximation, the stress distribution in thermally tempered glass can be represented by a simple parabola, with the magnitude of the surface compressive stress approximately equal to twice the center tension.

When thermally tempered glass breaks, unlike annealed glass, it shatters into rock-salt like pieces which do not have sharp edges or needle-like shapes. This behavior is particularly useful for situations where safe fracture behavior is necessary and it is for this reason that characterizing the fracture behavior of thermally tempered glass is of paramount importance. The desired fracture behavior is called “dicing” and occurs when the glass has achieved full temper. The dicing threshold of tempered glass is a somewhat arbitrarily defined fracture behavior which can be considered “safe” to the user in the event of glass failure. Standards for dicing thresholds exist worldwide, such as ASTM C1048 and ANSI Z97.1 in the United States, EN12150-1 in Europe, GOST 5727-88 in Russia, JIS R 3206 in Japan, and GB 15763.2 in China (all of which are hereby incorporated by reference). The standards across countries are similar in that they generally state a fragmented piece of tempered soda-lime glass is required to contain at least 30-40 fragments in an area of $50\text{ mm}\times 50\text{ mm}$ ($1.6\text{ fragments}/\text{cm}^2$) for thick glasses ($>3\text{ mm}$), while Japanese standards in particular require at least 60 fragments in the case of thinner glass.

The glasses disclosed herein have high coefficients of thermal expansion and can be used with a thermal tempering process to obtain improved fracture behavior when compared to commercially available thin glasses. The glasses described herein are needed to satisfy a growing demand for stronger but thinner thermally strengthened glasses for commercial electronics, automotive and architectural applications where durability and/or scratch resistance are desired along with a “safe” break pattern. Additionally, the glasses must also retain a significant chemical durability, as they will likely be exposed to the elements for extended periods of time.

It has been found herein that in order for thin glasses (3 mm or less, 2 mm or less, or 1 mm or less) to be thermally temperable and retain the desired fracture pattern, the low temperature coefficient of thermal expansion (LTCTE) should be $75\times 10^{-7}/^\circ\text{C}$. or greater and the high temperature coefficient of thermal expansion (HTCTE) should be $250\times 10^{-7}/^\circ\text{C}$. or greater. In some embodiments, it has been found that in addition to the LTCTE and the HTCTE limitations, the glass must also have a fracture toughness (K_{IC}) is less than $0.65\text{ MPa}\cdot\text{m}^{1/2}$.

In some embodiments, the glass comprises a combination of SiO_2 , K_2O , Al_2O_3 , B_2O_3 , and alkaline earth oxides. For example, embodiments may comprise greater than 70 mol % SiO_2 , greater than 0 mol % Al_2O_3 , greater than 0 mol % B_2O_3 , greater than 3 mol % K_2O , and greater than 0 mol % MO, where M is Ca, Ba, Sr, or Mg. In some embodiments, the glass may comprise greater than 72 mol % SiO_2 , >0 to 10 mol % Al_2O_3 , 0.5 to 5 mol % B_2O_3 , 3 to 20 mol % K_2O , and 2 to 20 mol % MO, where M is Ca, Ba, Sr, or Mg. In other embodiments, the glass may comprise 72 to 90 mol % SiO_2 , 3 to 10 mol % Al_2O_3 , 1 to 5 mol % B_2O_3 , 8 to 18 mol % K_2O , and 2 to 15 mol % MO, where M is Ca, Ba, Sr, or Mg.

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SiO₂, which is the largest oxide component of the embodied glasses, maybe included to provide high temperature stability and chemical durability. In some embodiments, the glass can comprise 70 mol % or greater SiO₂. In some embodiments, the glass can comprise 72 mol % or greater SiO₂. In some embodiments, the glass can comprise 72 to 90 mol % SiO₂. In some embodiments, the glass can comprise from 75 to 90 mol % SiO₂. In some embodiments, the glass can comprise 70 to 92 mol %, 70 to 90 mol %, 70 to 85 mol %, 70 to 80 mol %, 72 to 92 mol %, 72 to 90 mol %, 72 to 85 mol %, 72 to 80 mol %, 75 to 92 mol %, 75 to 90 mol %, 75 to 85 mol %, or 75 to 80 mol % SiO₂. In some embodiments, the glass comprises 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91 or 92 mol % SiO₂.

Al₂O₃ may influence the structure of the glass and, additionally, lower the liquidus temperature and coefficient of thermal expansion, or enhance the strain point. In some embodiments, the glass can comprise greater than 0 mol % Al₂O₃. In some embodiments, the glass can comprise from >0 to 20 mol % Al₂O₃. In some embodiments, the glass can comprise from >0 to 10 mol %, 3 to 10 mol %, or 5 to 10 mol % Al₂O₃. In some embodiments, the glass can comprise from >0 to 20 mol %, >0 to 18 mol %, >0 to 15 mol %, >0 to 12 mol %, >0 to 10 mol %, >0 to 8 mol %, 1 to 20 mol %, 1 to 18 mol %, 1 to 15 mol %, 1 to 12 mol %, 1 to 10 mol %, 1 to 8 mol %, 3 to 20 mol %, 3 to 18 mol %, 3 to 15 mol %, 3 to 12 mol %, 3 to 10 mol %, 3 to 8 mol %, 5 to 20 mol %, 5 to 18 mol %, 5 to 15 mol %, 5 to 12 mol %, 5 to 10 mol %, 5 to 8 mol %, 7 to 20 mol %, 7 to 18 mol %, 7 to 15 mol %, 7 to 12 mol %, or 7 to 10 mol % Al₂O₃. In some embodiments, the glass can comprise about >0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 mol % Al₂O₃.

Without being bound by theory, it is believed that incorporating B₂O₃ into the glasses described herein impacts the coefficient of thermal expansion, especially at high temperatures, and improves the temperability of the glasses. In some embodiments, when present, the glass can comprise >0 mol % to 10 mol % B₂O₃. In some embodiments, the glass can comprise from 0.5 to 10 mol %, 0.5 mol % to 8 mol % or from 1 mol % to 6 mol % B₂O₃. In some embodiments, the glass can comprise from 1 to 5 mol % B₂O₃. In some embodiments, the glass can comprise from >0 to 10 mol %, >0 to 8 mol %, >0 to 6 mol %, >0 to 5 mol %, >0 to 4 mol %, 0.5 to 10 mol %, 0.5 to 8 mol %, 0.5 to 6 mol %, 0.5 to 5 mol %, 0.5 to 4 mol %, 1 to 10 mol %, 1 to 8 mol %, 1 to 6 mol %, 1 to 5 mol %, 1 to 4 mol %, 2 to 10 mol %, 2 to 8 mol %, 2 to 6 mol %, 2 to 5 mol %, or 2 to 4 mol %. In some embodiments, the glass can comprise about 0, >0, 0.5, 1, 1.5, 2, 3, 4, 5, 6, 7, 8, 9, or 10 mol % B₂O₃.

Alkaline earth oxides may improve desirable properties in the materials, including influencing the Young's modulus and the coefficient of thermal expansion. In some embodiments, the glass comprises from >0 mol % to about 20 mol % MO (0 mol % < MO ≤ 20 mol %), where M is the sum of the alkaline earth metals Mg, Ca, Sr, and Ba, in the glass. In some embodiments, the glass can comprise from 2 to 20 mol % MO. In some embodiments, the glass can comprise from 2 to 15 mol % MO. In some embodiments, the glass can comprise about >0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 mol % MO.

In some embodiments, the glasses comprise MgO, CaO, or SrO. In some embodiments, the glass can comprise greater than 0 mol % MgO. In some embodiments, the glass can comprise from >0 to 10 mol % MgO. In some embodiments, the glass can comprise from 3 to 10 mol %, 5 to 10

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mol %, 5 to 8 mol % MgO. In some embodiments, the glass can comprise from >0 to 10 mol %, >0 to 8 mol %, >0 to 6 mol %, >0 to 4 mol %, >0 to 2 mol %, 1 to 10 mol %, 1 to 8 mol %, 1 to 6 mol %, 1 to 4 mol %, 1 to 2 mol %, 3 to 8 mol %, 3 to 6 mol %, 3 to 10 mol %, 5 to 8 mol %, 5 to 10 mol %, 7 to 10 mol %, or 8 to 10 mol % MgO. In some embodiments, the glass can comprise about >0, 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 mol % MgO.

In some embodiments, the glass can comprise greater than 0 mol % CaO. In some embodiments, the glass can comprise from >0 to 15 mol % CaO. In some embodiments, the glass can comprise from >0 to 5 mol %, 6 to 13 mol %, 5 to 15 mol %, 7 to 13 mol %, 7 to 11 mol %, 8 to 12 mol % CaO. In some embodiments, the glass can comprise from >0 to 15 mol %, >0 to 13 mol %, >0 to 11 mol %, >0 to 9 mol %, >0 to 7 mol %, >0 to 5 mol %, 1 to 15 mol %, 1 to 13 mol %, 1 to 11 mol %, 1 to 9 mol %, 1 to 7 mol %, 1 to 5 mol %, 3 to 15 mol %, 3 to 13 mol %, 3 to 11 mol %, 3 to 9 mol %, 3 to 7 mol %, 3 to 5 mol %, 5 to 15 mol %, 5 to 13 mol %, 5 to 11 mol %, 5 to 9 mol %, 5 to 7 mol %, 7 to 15 mol %, 7 to 13 mol %, 7 to 11 mol %, 7 to 9 mol %, 9 to 15 mol %, 9 to 13 mol %, 9 to 11 mol %, 11 to 15 mol %, or 11 to 13 mol % CaO. In some embodiments, the glass can comprise about >0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 mol % CaO.

SrO may be present in some embodiments and in such embodiments, the glass can comprise from 0 to 10 mol % SrO. In some embodiments, the glass can comprise from >0 to 10 mol % SrO. In some embodiments, the glass can comprise from 3 to 10 mol %, 5 to 10 mol %, 5 to 8 mol % SrO. In some embodiments, the glass can comprise from >0 to 10 mol %, >0 to 8 mol %, >0 to 6 mol %, >0 to 4 mol %, >0 to 2 mol %, 1 to 10 mol %, 1 to 8 mol %, 1 to 6 mol %, 1 to 4 mol %, 1 to 2 mol %, 3 to 8 mol %, 3 to 6 mol %, 3 to 10 mol %, 5 to 8 mol %, 5 to 10 mol %, 7 to 10 mol %, or 8 to 10 mol % SrO. In some embodiments, the glass can comprise about >0, 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 mol % SrO.

BaO may be present in some embodiments and in such embodiments, the glass can comprise from 0 to 15 mol % BaO. In some embodiments, the glass can comprise from 0 to 10 mol %, >0 to 5 mol %, 6 to 13 mol %, 5 to 15 mol %, 7 to 13 mol %, 7 to 11 mol %, 8 to 12 mol % BaO. In some embodiments, the glass can comprise from >0 to 15 mol %, >0 to 13 mol %, >0 to 11 mol %, >0 to 9 mol %, >0 to 7 mol %, >0 to 5 mol %, 1 to 15 mol %, 1 to 13 mol %, 1 to 11 mol %, 1 to 9 mol %, 1 to 7 mol %, 1 to 5 mol %, 3 to 15 mol %, 3 to 13 mol %, 3 to 11 mol %, 3 to 9 mol %, 3 to 7 mol %, 3 to 5 mol %, 5 to 15 mol %, 5 to 13 mol %, 5 to 11 mol %, 5 to 9 mol %, 5 to 7 mol %, 7 to 15 mol %, 7 to 13 mol %, 7 to 11 mol %, 7 to 9 mol %, 9 to 15 mol %, 9 to 13 mol %, 9 to 11 mol %, 11 to 15 mol %, or 11 to 13 mol % BaO. In some embodiments, the glass can comprise about >0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 mol % BaO.

Na₂O and K₂O may improve the temperability of the glass and influence the coefficient of thermal expansion, especially at low temperatures. In some embodiments, the glass can comprise from 0 to 10 mol % Na₂O. In some embodiments, the glass can comprise >0 to 10 mol % Na₂O. In some embodiments, the glass can comprise 0 to 8 mol % Na₂O. In some embodiments, the glass can comprise 2 to 6 mol % Na₂O. In some embodiments, the glass can comprise from 0 to 10 mol %, 0 to 8 mol %, 0 to 6 mol %, 0 to 4 mol %, 0 to 2 mol %, >0 to 10 mol %, >0 to 8 mol %, >0 to 6 mol %, >0 to 4 mol %, >0 to 2 mol %, 2 to 10 mol %, 2 to 8 mol %, 2 to 6 mol %, 2 to 4 mol %, 5 to 16 mol %, 5 to

10 mol %, 5 to 8 mol %, or 8 to 10 mol % Na₂O. In some embodiments, the glass can comprise 0, >0, 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 mol % Na₂O.

In some embodiments, the glass can comprise from 3 to 20 mol % K₂O. In some embodiments, the glass can comprise 5 to 18 mol % K₂O. In some embodiments, the glass can comprise 8 to 12 mol % K₂O. In some embodiments, the glass can comprise from 3 to 20 mol %, 3 to 18 mol %, 3 to 15 mol %, 3 to 12 mol %, 3 to 10 mol %, 3 to 8 mol %, 5 to 20 mol %, 5 to 18 mol %, 5 to 15 mol %, 5 to 12 mol %, 5 to 10 mol %, 5 to 8 mol %, 8 to 20 mol %, 8 to 18 mol %, 8 to 15 mol %, 8 to 12 mol %, 8 to 10 mol %, 10 to 20 mol %, 10 to 18 mol %, 10 to 15 mol %, 10 to 12 mol %, 12 to 20 mol %, 12 to 18 mol %, or 12 to 15 mol % K₂O. In some embodiments, the glass can comprise 0, >0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 mol % K₂O.

In some embodiments, Li₂O may be present and in such embodiments, the glass can comprise from 0 to 5 mol % Li₂O. In some embodiments, the glass can comprise from >0 to 5 mol % Li₂O. In some embodiments, the glass can comprise from about >0 to 3.5 mol % Li₂O or 0.2 to 3 mol % Li₂O. In some embodiments, the glass can comprise from 1 to 4 mol % Li₂O. In some embodiments, the glass can comprise from 0.2 to 5 mol %, 0.2 to 4 mol %, 0.2 to 3 mol %, 0.2 to 2 mol %, >0 to 5 mol %, >0 to 4 mol %, >0 to 3 mol %, >0 to 2 mol %, 1 to 5 mol %, 1 to 4 mol %, or 1 to 3 mol % Li₂O. In some embodiments, the glass can comprise about 0, >0, 1, 2, 3, 4, or 5 mol % Li₂O.

In some embodiments, the total amount of the alkalis Na₂O and K₂O or Li₂O, Na₂O, and K₂O is important to the glass properties. In some embodiments, the glass can comprise 3 to 22 mol % Na₂O+K₂O or Li₂O+Na₂O+K₂O. In some embodiments, the glass can comprise 5 to 20 mol % Na₂O+K₂O or Li₂O+Na₂O+K₂O. In some embodiments, the glass can comprise 5 to 15 mol % Na₂O+K₂O or Li₂O+Na₂O+K₂O. In some embodiments, the glass can comprise 8 to 16 mol % Na₂O+K₂O or Li₂O+Na₂O+K₂O. In some

embodiments, the glass can comprise 9 to 14 mol % Na₂O+K₂O or Li₂O+Na₂O+K₂O. In some embodiments, the glass can comprise from 3 to 22 mol %, 5 to 22 mol %, 8 to 22 mol %, 3 to 20 mol %, 5 to 20 mol %, 8 to 20 mol %, 3 to 15 mol %, 5 to 15 mol %, 8 to 15 mol %, 3 to 12 mol %, 5 to 12 mol %, or 8 to 12 mol % Na₂O+K₂O or Li₂O+Na₂O+K₂O. In some embodiments, the glass can comprise 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, or 22 mol % Na₂O+K₂O or Li₂O+Na₂O+K₂O.

Additional components can be incorporated into the glass to provide additional benefits or may be incorporated as contaminants typically found in commercially-prepared glass. For example, additional components can be added as coloring or fining agents (e.g., to facilitate removal of gaseous inclusions from melted batch materials used to produce the glass) and/or for other purposes. In some embodiments, the glass may comprise one or more compounds useful as ultraviolet radiation absorbers. In some embodiments, the glass can comprise 3 mol % or less ZnO, TiO₂, CeO, MnO, Nb₂O₅, MoO₃, Ta₂O₅, WO₃, SnO₂, Fe₂O₃, As₂O₃, Sb₂O₃, Cl, Br, or combinations thereof. In some embodiments, the glass can comprise from 0 to about 3 mol %, 0 to about 2 mol %, 0 to about 1 mol %, 0 to 0.5 mol %, 0 to 0.1 mol %, 0 to 0.05 mol %, or 0 to 0.01 mol % ZnO, TiO₂, CeO, MnO, Nb₂O₅, MoO₃, Ta₂O₅, WO₃, SnO₂, Fe₂O₃, As₂O₃, Sb₂O₃, Cl, Br, or combinations thereof. The glasses, according to some embodiments, can also include various contaminants associated with batch materials and/or introduced into the glass by the melting, fining, and/or forming equipment used to produce the glass. For example, in some embodiments, the glass can comprise from 0 to about 3 mol %, 0 to about 2 mol %, 0 to about 1 mol %, 0 to about 0.5 mol %, 0 to about 0.1 mol %, 0 to about 0.05 mol %, or 0 to about 0.01 mol % SnO₂ or Fe₂O₃, or combinations thereof.

Non-limiting examples of amounts of precursor oxides for forming the embodied glasses are listed in Table 1, along with the properties of the resulting glasses.

TABLE 1

Analyzed mol %	Glass A	Glass B	Glass C	Glass D	Glass E
SiO ₂	76.60	75.29	75.11	75.75	76.10
B ₂ O ₃	1.88	2.35	2.57	2.15	2.41
Al ₂ O ₃	6.32	7.06	7.00	6.07	6.03
K ₂ O	11.61	12.32	11.41	11.08	11.11
CaO	0.00	0.00	0.00	0.00	0.00
SrO	3.84	2.93	3.86	4.89	3.87
P ₂ O ₅	0.00	0.00	0.00	0.00	0.42
SnO ₂	0.05	0.05	0.05	0.05	0.05
LTCTE (ppm/° C.)	7.87	7.80	7.71	7.69	—
HTCTE (ppm/° C.)	—	27.31	28.54	27.12	—
K _{TC} (MPa·m ^{1/2})	0.55	—	0.65	0.62	—
Density (g/cm ³)	2.471	2.456	2.471	2.492	—
Strain Pt (° C.)	600	607	610	611	—
Anneal Pt (° C.)	652	658	662	662	—
Softening Pt (° C.)	876	889	890	882	—
SOC (TPa ⁻¹)	2.965	3.121	3.049	3.015	—
Refractive Index	1.5005	1.5005	1.5019	1.5038	—
200 Poise temperature (° C.)	1720	1762	1758	1706	1762
35000 Poise temperature (° C.)	1172	1190	1204	1181	1195
200000 Poise temperature (° C.)	1072	1085	1099	1080	1107
Analyzed mol %	Glass F	Glass G	Glass H	Glass I	Glass J
SiO ₂	75.64	76.00	75.83	75.35	75.35
B ₂ O ₃	2.25	2.10	2.29	2.34	1.58
Al ₂ O ₃	6.05	6.04	6.02	6.00	6.00

TABLE 1-continued

K ₂ O	12.01	11.94	11.81	12.18	13.18
CaO	4.01	0.00	0.00	0.00	0.00
SrO	0.00	3.84	3.89	3.86	3.85
P ₂ O ₅	0.00	0.04	0.11	0.22	0.00
SnO ₂	0.05	0.05	0.05	0.05	0.05
LTCTE (ppm/° C.)	7.82	7.85	7.78	7.91	8.33
HTCTE (ppm/° C.)	27.68	24.72	27.08	26.88	27.21
K _{TC} (MPa-m ^{1/2})	0.59	0.67	0.62	—	0.63
Density (g/cm ³)	2.415	2.476	2.475	2.476	2.480
Strain Pt (° C.)	615	598	602	599	586
Anneal Pt (° C.)	667	649	654	649	638
Softening Pt (° C.)	887	872.2	870.8	875.3	858.3
SOC (TPa ⁻¹)	3.050	2.976	3.011	2.967	2.951
Refractive Index	1.5012	1.5029	1.5022	1.5018	1.5030
200 Poise temperature (° C.)	1721	1695	1715	1684	1684
35000 Poise temperature (° C.)	1185	1162	1162	1157	1148
200000 Poise temperature (° C.)	1084	1062	1063	1057	1049

As noted above, the embodied glasses of one or more embodiments may exhibit a color. In some embodiments, it is desirable for the glass to have a green-yellow to gold to amber color. In particular embodiments, these colors may be combined with the blue color of a crystalline silicon photovoltaic cell to create a dark blue to black color that is aesthetically pleasing. In some embodiments, the glasses exhibit a color presented in SCE color space coordinates (determined from reflectance spectra measurements using a spectrophotometer, with illuminant D65 and specular reflectance excluded), with the following ranges: a*=from about -10 to about 30; b*=from about 0 to about 30; and L*>80 for a glass having a thickness of 2.7 mm. In some embodiments, the glasses exhibit a color presented in SCI color space coordinates of a*=from about -5 to about -1; b*=from about 5 to about 18; and L*>83 for a glass having a thickness of 2.7 mm. In some applications, the combination of the photovoltaic cell and the glass combine to produce a desired color. In such application, the SCE color space coordinates (determined from reflectance spectra measurements using a spectrophotometer, with illuminant D65 and specular reflectance excluded) of the combination of the glass and the photovoltaic cell may comprise the following ranges: a*=from about 0 to about 10; b*=from about -15 to about 0; and L* from about 10 to about 50.

In some embodiments, the glass can be strengthened via thermal or chemical tempering. In some embodiments, the glass can be strengthened to include compressive stress (CS) that extends from a surface thereof to a depth of compression (DOC). The compressive stress regions are balanced by a central portion exhibiting a tensile stress. At the DOC, the stress crosses from a positive (compressive) stress to a negative (tensile) stress. Such strengthening methods comprise both thermal tempering and chemical tempering.

The glasses described herein are particularly capable of being thermally tempered. The process of thermal tempering is well known in the art. To thermally temper a glass article, the article is heated to near the softening temperature and then rapidly cooled or quenched. As a result, the glass will possess a lower surface temperature than the interior during cooling. The temperature difference is maintained until the surface of the glass cools to at least its strain point or lower, such as room temperature. Because the center of the glass cools more slowly, it contracts to a smaller specific volume while the high specific volume of the surface layer remains unchanged. This leads to a surface compressive layer that

gives tempered glass its strength. The difference in specific volume is, in part, due to differences in the thermal expansion of the glass upon cooling, while to a lesser extent from a fictive temperature difference between the surface and the bulk. To a first approximation, the stress distribution in thermally tempered glass can be represented by a simple parabola, with the magnitude of the surface compressive stress approximately equal to twice the center tension.

As an alternative to thermal tempering, the glasses disclosed herein may be ion exchanged by immersion in at least one ion exchange bath containing molten salts (e.g., nitrates, sulfides, halides, or the like) of at least one alkali metal such as lithium, sodium, or potassium. Ion exchange is commonly used to chemically strengthen glasses. In one particular example, alkali cations within a source of such cations (e.g., a molten salt, or "ion exchange," bath) are exchanged with smaller alkali cations within the glass to achieve a layer under a compressive stress (CS) extending from the surface of the glass to a depth of compression (DOC) within the glass phase. For example, potassium ions from the cation source are often exchanged with sodium and/or lithium ions within the glass phase, and the K⁺ concentration profile correlates with the compressive stress and depth of layer. The ion exchange bath may contain a salt (or salts) of a single alkali metal (e.g., sulfides, nitrates, or halides of Li, Na, or K) or salts of two or more alkali metals (e.g., sulfides, nitrates, or halides of Li and Na, or sulfides, nitrates, or halides of Na and K). Ion exchange is carried out in the ion exchange bath at temperatures ranging from about 390° C. to about 550° C. for times ranging from about 0.5 hour to about 24 hours.

The glass, in some embodiments, is ion exchanged and has a compressive layer extending from a surface to a depth of compression (DOC) of at least about 10 μm or, in some embodiments, at least about 30 μm into the glass, or in some embodiments up to about 10, 15, 20 or 25% into the glass as measured by thickness (surface to center). In some embodiments, the compressive layer extends from the surface of the glass to a depth of up to about 20% of the thickness of the glass. In some embodiments, the glass may be strengthened to exhibit a surface compressive stress in a range from 250 MPa to 800 MPa or greater.

In the strengthened glass, the depth of the compressive layer may be determined by electron microprobe, glow-discharge optical emission spectroscopy (GDOES, which is a technique for measuring depth profiles of constituent

elements in a solid sample by detecting emissions from atoms accommodated in plasma by sputtering), or similar techniques that can provide composition data as a function of depth, where data would show incorporation of Na (where Na⁺ replaces Li⁺ in the glass phase) and/or K at the surfaces. The DOC of a precursor glass may be measured by surface stress meter (FSM) using commercially available instruments such as the FSM-6000, manufactured by Orihara Industrial Co., Ltd. (Japan). Surface stress measurements rely upon the accurate measurement of the stress optical coefficient (SOC), which is related to the birefringence of the glass. SOC in turn is measured by those methods that are known in the art, such as fiber and four point bend methods, both of which are described in ASTM standard C770-98 (2013), entitled "Standard Test Method for Measurement of Glass Stress-Optical Coefficient," the contents of which are incorporated herein by reference in their entirety, and a bulk cylinder method. CS may also be measured by measured by FSM. As used herein CS may be the "maximum compressive stress" which is the highest compressive stress value measured within the compressive stress layer. In some embodiments, the maximum compressive stress is located at the surface of the glass. In other embodiments, the maximum compressive stress may occur at a depth below the surface, giving the compressive profile the appearance of a "buried peak."

The thermally or chemically strengthened glasses or articles disclosed herein may be incorporated into another article such as an article with a display (or display articles) (e.g., consumer electronics, including mobile phones, tablets, computers, navigation systems, and the like), architectural articles (e.g., windows, skylights, shingles), transportation articles (e.g., automotive, trains, aircraft, sea craft, etc.), appliance articles, or any article that would benefit from transparency, scratch-resistance, abrasion resistance or a combination thereof. In other embodiments, the glass forms a portion of a consumer electronic product, such as a cellular phone or smart phone, laptop computer, tablet, or the like. Such consumer electronic products typically comprise a housing having front, back, and side surfaces, and include electrical components such as a power source, a controller, a memory, a display, and the like, which are at least partially internal to the housing. In some embodiments, the glass described herein comprises at least a portion of a protective element, such as, but not limited to, the housing and/or display of a consumer electronic product.

Processes for Making Glasses

Glasses having the oxide contents listed in Table 1 can be made via traditional methods. For example, in some embodiments, the precursor glasses can be formed by thoroughly mixing the requisite batch materials (for example, using a turbular mixer) in order to secure a homogeneous melt, and subsequently placing into silica and/or platinum crucibles. The crucibles can be placed into a furnace and the glass batch melted and maintained at temperatures ranging from 1250-1650° C. for times ranging from about 6-16 hours. The melts can thereafter be poured into steel molds to yield glass slabs. Subsequently, those slabs can be transferred immediately to an annealer operating at about 500-650° C., where the glass is held at temperature for about 1 hour and subsequently cooled overnight. In another non-limiting example, precursor glasses are prepared by dry blending the appropriate oxides and mineral sources for a time sufficient to thoroughly mix the ingredients. The glasses are melted in platinum crucibles at temperatures ranging from about

1100° C. to about 1650° C. and held at temperature for about 16 hours. The resulting glass melts are then poured onto a steel table to cool. The precursor glasses are then annealed at appropriate temperatures.

Tempering of the embodied glasses was achieved using conventional processes wherein the glasses were heated in a radiant energy furnace or a convection furnace (or a "combined mode" furnace using both techniques) to a predetermined temperature, then gas cooling ("quenching"), typically via convection by blowing large amounts of ambient air against or along the glass surface.

While typical embodiments have been set forth for the purpose of illustration, the foregoing description should not be deemed to be a limitation on the scope of the disclosure or appended claims. Accordingly, various modifications, adaptations, and alternatives may occur to one skilled in the art without departing from the spirit and scope of the present disclosure or appended claims.

The invention claimed is:

1. A method of making a glass article, comprising: mixing and melting batch constituents comprising:
 - from 75 to 90 mol % SiO₂;
 - from 7 to 20 mol % Al₂O₃;
 - greater than 0 to 20 mol % of at least one alkaline earth oxide selected from the group consisting of MgO, CaO, BaO, or SrO;
 - from 8 to 20 mol % K₂O;
 - greater than 0 to 10 mol % B₂O₃, and
 - 0 mol % Na₂O to form a melt of the glass,
 wherein a low temperature coefficient of thermal expansion from 25° C. to 300° C. of the glass is greater than 75×10⁻⁷ ppm/° C., and
 wherein a fracture toughness, K_{IC} of the glass is less than 0.65 MP·m^{1/2}.
2. The method of claim 1, further comprising forming the melt into glass sheets.
3. The method of claim 2, further comprising thermally tempering the glass sheets.
4. The method of claim 3, further comprising maintaining the melt at temperatures ranging from 1100-1650° C. for times ranging from about 6-16 hours, wherein the glass sheets meet dicing standard ASTM C1048 after the thermal tempering.
5. The method of claim 4, further comprising annealing the glass sheets at about 500-650° C.
6. The method of claim 5, wherein during the annealing the glass is held at temperature for about 1 hour and subsequently cooled for at least 6 hours.
7. The method of claim 1, wherein a high temperature coefficient of thermal expansion of the glass is greater than 250×10⁻⁷ ppm/° C.
8. The method of claim 1, wherein the glass has a viscosity of 200 kP at a temperature within a range of 1000-1200° C.
9. The method of claim 1, wherein the glass has a viscosity of 35 kP at a temperature within a range of 1100-1300° C.
10. The method of claim 1, wherein the mixing comprises mixing from 0.5 to 8 mol % B₂O₃.
11. The method of claim 1, wherein the mixing comprises mixing from 2 to 20 mol % MgO.
12. The method of claim 1, wherein the mixing comprises mixing from 11 to 20 mol % K₂O.
13. The method of claim 1, wherein the glass has a surface compressive stress of at least 250 MPa.
14. The method of claim 1, wherein the glass meets the dicing standard ASTM C1048.

15. The method of claim **3**, wherein the glass has a depth of compression of at least about 10 microns, and the depth of compression extends up to about 25% into the glass as measured by thickness from surface to a center of the glass.

16. The method of claim **1**, wherein the mixing comprises 5 mixing:

from 7 to 17 mol % Al_2O_3 ;

from 0 to 10 mol % of the at least one alkaline earth oxide; and

from 8 to 18 mol % K_2O . 10

17. The method of claim **1**, wherein the mixing comprises mixing 0 mol % CaO .

18. A method of making a glass article, comprising: mixing and melting batch constituents comprising:

from 75 to 90 mol % SiO_2 ; 15

from 7 to 20 mol % Al_2O_3 ;

greater than 0 to 20 mol % of at least one alkaline earth oxide selected from the group consisting of MgO , CaO , BaO , or SrO ;

from 8 to 20 mol % K_2O ; and 20

greater than 0 to 2.57 mol % B_2O_3 to form a melt of the glass,

wherein the glass exhibits a 200 Poise temperature that is within a range of from 1684° C. to 1762° C.

19. The method of claim **1**, wherein 25 the mixing comprises mixing from greater than 0 to 2.57 mol % B_2O_3 .

20. The method of claim **18**, wherein 30 the mixing comprises mixing from 12 to 20 mol % K_2O , and

the glass exhibits a 200 Poise temperature that is within a range of 1700° C. to 1762° C.

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